

WASP-49b

R-1112

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_190209 · created 2026-07-30T20:57:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458523.6772 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.04
Transit depth (Rp/Rs)^2	1.48 +/- 0.97 [%]
Orbital Inclination (inc)	83.91 +/- 2.9
Airmass coefficient 1 (a1)	1.036 +/- 0.023
Airmass coefficient 2 (a2)	-0.019 +/- 0.048
Scatter in the residuals of the lightcurve fit is	2.21 %
Best Comparison Star	#3 - [566, 67]
Optimal Aperture	3.93
Optimal Annulus	6.61
Transit Duration (day)	0.068 +/- 0.044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.04 vs 0.116 (deviation 0.15σ)
Duration fit vs published	0.068 d vs 0.09 d (deviation 0.5σ)
Mid-time O–C	6.4 min
Depth SNR	3.0
Residual scatter	2.21 %

Light curve

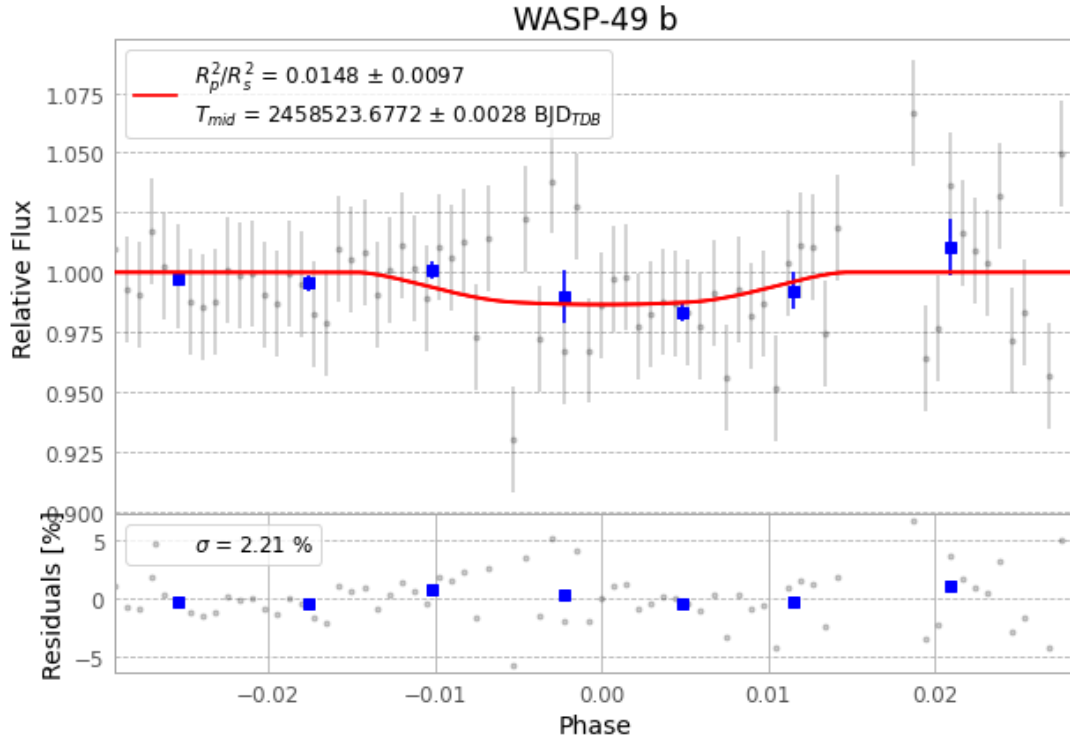


Figure 1 — FinalLightCurve_WASP-49 b_2019-02-08.png (SHA-256 c794c0c160d972da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [535, 160], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fcf7c8fe2cba241d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), WASP-49190209021212.FITS → WASP-49190209060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-190209.log (fbf8317a88eb...), FinalLightCurve_WASP-49 b_2019-02-08.png (c794c0c160d9...), FinalLightCurve_WASP-49 b_2019-02-08.csv (2868d0874f9c...)

Compute resources

Wall time 335.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:57Z.